

PROGRAM:

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#include <stdio.h>
#define MAX_SIZE 100 // Maximum size of the array
void createArray(int arr[], int *size);
void insertElement(int arr[], int *size, int element, int position);
int searchElement(int arr[], int size, int element);
void deleteElement(int arr[], int *size, int position);
void displayArray(int arr[], int size);
int main() {
    int arr[MAX_SIZE];
    int size = 0;
    int choice, element, position, result;
    while(1) {
        printf("\nArray Operations Menu:\n");
        printf("1. Create Array\n");
        printf("2. Insert Element\n");
        printf("3. Search Element\n");
        printf("4. Delete Element\n");
        printf("5. Display Array\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                createArray(arr, &size);
                break;
            case 2:
                printf("Enter element to insert: ");
                scanf("%d", &element);
                printf("Enter position to insert (0 to %d): ", size);
                scanf("%d", &position);
                insertElement(arr, &size, element, position);
                break;
            case 3:
                printf("Enter element to search: ");
                scanf("%d", &element);
                result = searchElement(arr, size, element);
                if(result != -1)
                    printf("Element found at position: %d\n", result);
                else
                    printf("Element not found in the array.\n");
                break;
            case 4:
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printf("Enter position to delete (0 to %d): ", size - 1);
scanf("%d", &position);
deleteElement(arr, &size, position);
break;
case 5:
displayArray(arr, size);
break;
case 6:
return 0;
default:
printf("Invalid choice! Please enter a valid option.\n");
}
}
return 0;
}

void createArray(int arr[], int *size) {
int n, i;
printf("Enter the number of elements: ");
scanf("%d", &n);
if(n > MAX_SIZE) {
printf("Error: Number of elements exceeds the maximum array size.\n");
return;
}
printf("Enter %d elements: ", n);
for(i = 0; i < n; i++) {
scanf("%d", &arr[i]);
}
*size = n;
printf("Array created successfully.\n");
}

void insertElement(int arr[], int *size, int element, int position) {
if(*size >= MAX_SIZE) {
printf("Error: Array is full. Cannot insert element.\n");
return;
}
if(position < 0 || position > *size) {
printf("Error: Invalid position.\n");
return;
}
for(int i = *size; i > position; i--) {
arr[i] = arr[i - 1];
}
arr[position] = element;
(*size)++;
}

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printf("Element inserted successfully.\n");
}
int searchElement(int arr[], int size, int element) {
for(int i = 0; i < size; i++) {
if(arr[i] == element) {
return i;
}
}
return -1;
}
void deleteElement(int arr[], int *size, int position) {
if(position < 0 || position >= *size) {
printf("Error: Invalid position.\n");
return;
}
for(int i = position; i < *size - 1; i++) {
arr[i] = arr[i + 1];
}
(*size)--;
printf("Element deleted successfully.\n");
}
void displayArray(int arr[], int size) {
if(size == 0) {
printf("Array is empty.\n");
return;
}
printf("Array elements: ");
for(int i = 0; i < size; i++) {
printf("%d ", arr[i]);
}
printf("\n");
}

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